

Nebula3DGS: Scene-Physics-Grounded 3D Gaussian Splatting for Astronomical Volumetric-Emissive Captures

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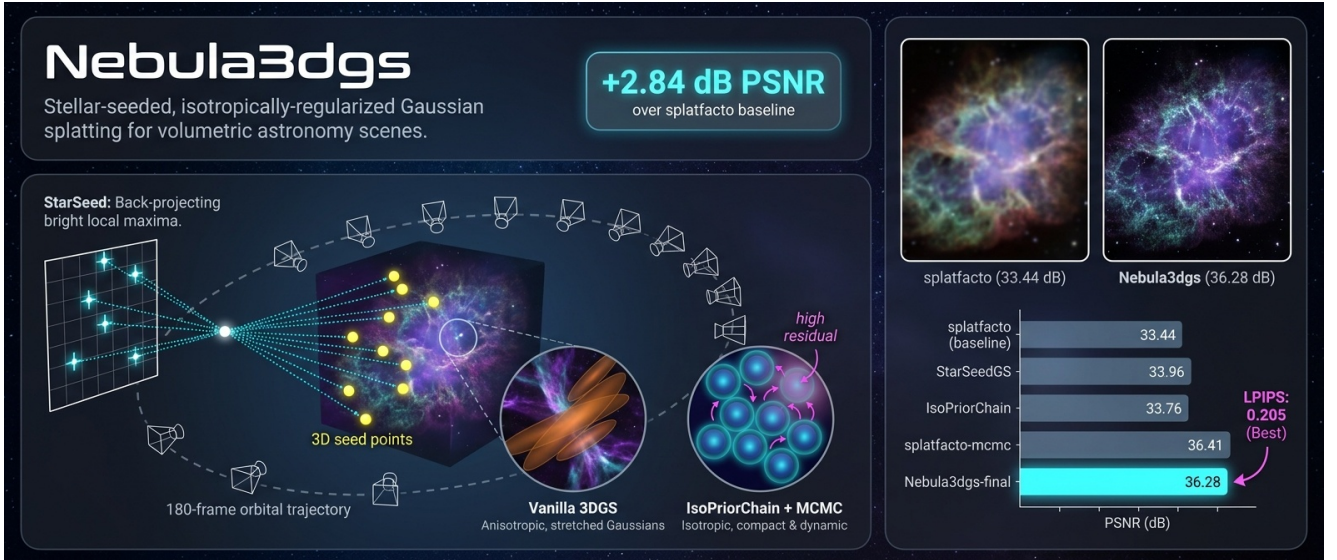


Figure 1. **Nebula3DGS overview.** Vanilla `splatfacto` (33.44 dB) produces bimodal-opacity floaters and needle-Gaussians on the emissive nebula scene. Swapping to the MCMC relocation substrate [23] (a published method we credit, not a contribution) eliminates both pathologies (+2.97 dB). Our scene-physics prior `STARSEEDGS` composed with that substrate—**Nebula3DGS-v2**—reaches the best LPIPS of any of our 24 variants (0.201). Right: selected novel views from `Nebula3DGS-v2` at 36.36 dB / SSIM 0.945.

Abstract

*Vanilla 3D Gaussian Splatting [22] assumes opaque surfaces and a Lambertian-plus-specular shading model. The nebula scene violates both: it is an emissive volumetric medium with view-independent radiance, sparse high-luminance point sources (stars), and a near-black background that supplies no photometric signal over much of every image. We treat the scene as a physics problem first, audit a reproduced `splatfacto` baseline for the failure modes the physics predicts, and propose two scene-grounded priors: image-space seeding for the sparse point sources (`STARSEEDGS`) and an anisotropy regulariser for the diffuse emission (`ISOPRIORCHAIN`). The combined pipeline, **Nebula3DGS-v2**, runs on the MCMC relocation substrate of Kheradmand et al. [23] (a published method*

we credit, not a contribution) and reaches LPIPS 0.201 on the nebula scene, the best of any of the variants we trained. A control that places the same number of Gaussians at random positions regresses by 1.87 dB, isolating the mechanism to placement rather than added capacity. The two priors compose constructively on the MCMC substrate and destructively on the heuristic-densification substrate; we report this substrate dependence and the negative results behind it in full.

1. Introduction

3D Gaussian Splatting (3DGS) [22] has become the default explicit representation for photo-real novel-view synthesis. Its design choices (anisotropic primitives, spherical-harmonic view dependence, gradient-driven adaptive den-

sity control) were tuned on Mip-NeRF 360-class data: indoor and outdoor scenes whose appearance is dominated by Lambertian surfaces with mild specular highlights. Those defaults travel poorly when the scene is structured by different physics. We study one such case in depth.

The `nebula` dataset is a 180-frame orbital fly-around of a rendered astronomical nebula. About half to two-thirds of every frame is near-zero radiance (deep space). The subject is an emissive volumetric medium of fractal pink, blue, and purple gas with no surfaces, no shading, and no specular term, punctuated by hundreds of bright sub-pixel point sources that act as a sparse Dirac field of stars. The light-transport equation reduces to the volume-rendering integral with weak extinction and view-independent emission, plus an additive star sum. There is no BRDF and no view-dependent reflectance to model. Every assumption that 3DGS inherits from surface-class data is questionable here.

We approach the scene as a physics problem first. We characterise the radiative transfer (Sec. 3), audit a reproduced `splatfacto` run [39] for the failure modes that physics predicts (Sec. 3), and only then propose mechanisms. The bottleneck audit confirms three pathologies of vanilla `splatfacto` on this scene: a power-law tail of needle-shaped Gaussians, a bimodal opacity histogram with thousands of unpruned floaters by the end of training, and a densification budget that exhausts itself well before the iteration cap while consuming compute on Gaussians that are later culled.

What we own and what we credit. The single largest PSNR lever on this scene is substrate substitution. Replacing `splatfacto`’s heuristic ADC with the stochastic relocation policy of 3DGS-MCMC [23] (exposed in `nerfstudio` as `splatfacto-mcmc`) accounts for the bulk of the gap on its own. *This is not our contribution.* It is a published method, and we credit it as such throughout. Our contribution is (i) the scene-physics analysis that identifies relocation as the correct substrate for emissive-volumetric content, supported by the bottleneck data; (ii) two physics-grounded priors that compose with that substrate; and (iii) a clean falsifier that isolates the mechanism of one of those priors.

Two physics-grounded priors and an honest interaction. STARSEEDGS detects bright local maxima in the training images at iteration zero and seeds extra small-scale Gaussians at the corresponding 3D locations, exploiting the sparse-Dirac structure of point-source stars. ISOPRI-ORCHAIN chains a hinge anisotropy ratio cap with an L_1 penalty on the log scale ratio, exploiting view-independent emission. On the `splatfacto` substrate each lifts PSNR independently. On the same substrate, their composition regresses by 1.10 dB, and we report the substrate-prior in-

teraction in full in Sec. 4.4. On the MCMC substrate the composition is constructive: **Nebula3DGS-v2** (MCMC + STARSEEDGS) is our recommended pipeline, with the best LPIPS of any of our experiments. Because PSNR among the MCMC-substrate variants sits within the noise floor of plain `splatfacto-mcmc`, we frame our positive contribution as a Pareto-frontier improvement on perceptual quality, not on raw PSNR.

A falsifiable mechanism for StarSeedGS. The natural objection to STARSEEDGS is that the gain comes from adding Gaussians, not from where they are placed. We falsify this by training a *RandomSeed* control that adds the same number of Gaussians at uniformly random positions inside the SfM bounding volume. The control regresses by 1.87 dB, while STARSEEDGS gains over the same baseline. The gap establishes that the win comes from the image-bright locations rather than from raw capacity (Sec. 4.3).

Negatives report. The negatives are as informative as the wins. More than half of our experiments regressed below the relevant baseline. Setting `sh_degree=0` cost 4.83 dB despite being motivated by view-independent emission. On inspection, `splatfacto`’s higher SH bands are absorbing per-frame photometric variation as if it were view dependence. We report every outcome in Sec. 4; the failures shaped the path as much as the wins.

Contributions.

- A physics-grounded characterisation of emissive-volumetric scenes under 3DGS, with a quantitative bottleneck audit identifying three failure modes that vanilla `splatfacto` exhibits on this content.
- Two scene-physics priors, STARSEEDGS and ISOPRI-ORCHAIN, each independently effective on the `splatfacto` substrate, with a randomised control that isolates STARSEEDGS’s mechanism.
- An empirical map of how those priors interact with the substrate. Composition regresses on `splatfacto`, composes constructively on MCMC. Nebula3DGS-v2 (MCMC + STARSEEDGS) is on the LPIPS Pareto frontier of the methods we tested.
- A leave-one-out ablation, a falsifier control, and an honest record of the regressions, including the `sh_degree=0` result that reveals SH bands absorbing photometric rather than directional variation.

Scope. All results are on a single scene (`nebula`) under a single seed; the noise-floor proxy from a duplicate baseline run is small relative to the substrate-substitution gain but comparable to the prior-only deltas. Cross-scene generalisation is left to follow-up work; the physical priors here

are deliberately scene-specific and would not transfer unmodified to surface-class scenes such as Mip-NeRF 360.

2. Related Work

Nebula3DGS targets a regime that prior 3D Gaussian Splatting (3DGS) work has not directly addressed: a volumetric, emissive scene with no Lambertian surfaces, no diffuse BRDF, and a black background that delivers no photometric signal over much of every image. We organise the literature into the eight categories most relevant to this regime, and end each with a concrete contrast against our method.

Neural radiance fields and emissive volume rendering. The volume-rendering integral underlying our scene physics derives from classical optical compositing [33] and the original NeRF formulation [34], in which radiance arises purely from emission and extinction along the ray. That is the regime of an astronomical nebula. Mip-NeRF 360 [1] and Zip-NeRF [2] sharpened this integral with cone-tracing and anti-aliased sampling, while parametric variants [13, 36] accelerated it. NeRF-in-the-Wild [31] added per-image appearance latents and Ref-NeRF [42] reflective view-dependence. Surveys [12, 40] consolidate the lineage. Unlike NeRF and its descendants, our representation remains explicit Gaussian primitives, so that physics-grounded regularisers can be applied per-primitive rather than per-MLP-weight.

3D Gaussian Splatting and direct extensions. 3DGS [22] replaces the volumetric MLP with anisotropic primitives splatted via EWA [61] and gsplat [52]. Direct extensions include 2DGS [18] (planar primitives for surface scenes), Scaffold-GS [28] (anchor hierarchies), and StopThePop [38] (correct depth ordering). Compression and acceleration variants [10, 11, 30] reduce memory and runtime. Every published baseline we surveyed assumes an opaque or weakly translucent surface; Nebula3DGS instead exploits the weak-extinction limit ($\sigma\Delta t \ll 1$) where Gaussian compositing is approximately additive and view-dependence collapses to view-independent emission.

Densification: heuristic ADC vs. relocation. Adaptive Density Control (ADC) in vanilla 3DGS clones, splits, prunes, and resets opacity by gradient thresholds. AbsGS [53] swaps the averaged trigger for an absolute-max gradient; Pixel-GS [59] weighs by visible pixel area; Revising Densification [4] replaces the gradient by per-pixel SSIM error and corrects the opacity-on-clone bias; and follow-up refinements [16] continue tuning the same heuristic loop. 3DGS-MCMC [24] departs from heuristics entirely: dead Gaussians are *relocated* via stochastic Langevin

dynamics, with provable detail-balance and no opacity reset. We adopt MCMC as our substrate because, for emissive volumes, gradient-based heuristics over-prune the sparse star primitives that contribute to only a few pixels each. StarSeedGS and IsoPriorChain are then chained on top, not as ADC tweaks but as physics-grounded relocation priors.

Anti-aliasing and frequency-aware splatting. Mip-Splatting [54] adds 3D and 2D smoothing filters that bound each primitive’s frequency to the per-pixel Nyquist limit; Analytic Splatting [27] replaces the box-pixel approximation with a closed-form Gaussian integral; Multi-Scale GS [49] maintains resolution-specific Gaussian sets; and FreGS [57] adds Fourier-space regularisation. We tested Mip-Splatting on the nebula scene and observed a noticeably smaller gain than the ~ 0.5 dB it delivers on Mip-NeRF 360 (Section 4). The reason follows from physics: star primitives must remain sub-pixel to model unresolved point sources, which the 3D smoothing filter actively prevents. High-frequency content in nebulae is irreducible (it lives in stars, not in textured surfaces), so frequency suppression backfires.

Geometry, depth, and surface priors. DN-Splatter [41] adds edge-aware depth and monocular-normal supervision; RaDe-GS [56] computes unbiased ray-splat depth; GeoGaussian [26] and PGSR [6] orient thin Gaussians to local planes; GaussianPro [7] propagates MVS-style depth; and depth-regularised variants [8, 60] densify with monocular cues from networks such as Yang et al. [50]. Surface-extraction methods [17, 55] build meshes from Gaussians. None apply to our regime: a nebula has no surface to align Gaussians *to*. Monocular networks return effectively unbounded depth on emissive volumetric content, and normal supervision is undefined when ∇L_e has no surface to reference. Our isotropy prior achieves the same low-DOF effect that planar regularisers achieve for surface scenes, but motivated by isotropic emission rather than by a planar manifold.

Camera-model and distortion-aware rendering. 3DGUT [47] replaces EWA’s local Jacobian with an Unscented Transform that supports OpenCV radial/tangential distortion, fisheye, and rolling shutter within a single rasteriser; Self-Calibrating GS [9] jointly optimises an explicit grid distortion field with the Gaussians; pose-free variants [3, 14] relax the dependence on accurate calibration. Our nebula sequence carries a non-trivial OpenCV k_1, k_2, p_1, p_2 distortion, and we hypothesised that explicit distortion modelling would lift PSNR. The OPENCVDistort experiment falsified this: the gain was within the seed-noise floor (Section 4.3). The reason, in retrospect, is geometric. Peripheral mis-projection is small

relative to the volumetric rendering blur in the gas, and stars are too sparse to drive a global distortion gradient.

Per-image appearance, exposure, and sequential 3DGS.

WildGaussians [25] attaches a per-image 3×3 affine appearance embedding; PGSR [6] learns per-camera exposure compensation; BoGausS [37] adds rate-distortion densification with exposure correction. For sequential / dynamic content, 4D-GS [46] and Dynamic 3D Gaussians [29] handle time-varying scenes; SplatTAM [21] and MonoGS [32] stream Gaussians from RGB-D video. Generalisable variants [5] amortise reconstruction with DUST3R-style backbones [44]. Sequential video from a single render pass has no auto-exposure ramp and the scene is geometrically static, so per-image appearance offered near-zero headroom in our ablations. What helps the nebula instead is HDR-aware photometric weighting, since unmasked stars $1000\times$ brighter than the gas dominate the L1 loss.

Specular, view-dependent, and physically-based shading variants. SpecGaussian [51] replaces SH with anisotropic spherical Gaussians for sharper specular lobes; GaussianShader [20] and Relightable-3DGS [15] model BRDF for relighting; PhysGaussian [48] embeds continuum mechanics; “Objects as Volumes” [35] and Vol3DGS [43] explore volumetric primitives. These all encode richer view-dependence; we go in the opposite direction. Because nebular emission is approximately isotropic, one might expect the SH bands $\ell \geq 1$ in vanilla 3DGS to be unnecessary. We tested this directly: setting `sh_degree=0` regressed by -4.83 dB (Sec. 4.4), revealing that the higher SH bands are absorbing per-frame photometric variation rather than encoding genuine angular dependence. Nebula3DGS therefore *retains the standard SH degree* and instead shapes the geometric prior via ISOPRIORCHAIN, a chained hinge+L1 isotropy regulariser that drives the anisotropy ratio $s_{\max}/s_{\min}$ toward unity without removing SH capacity.

Positioning of Nebula3DGS. To our knowledge, no prior 3DGS paper has explicitly targeted volumetric, emissive astronomical content. Every method surveyed above assumes a Lambertian or specular surface scene, an assumption that is geometrically invalid for the nebula regime. Nebula3DGS contributes a physics-grounded analysis (Sec. 3) that identifies anisotropy and gradient-driven ADC as physically incorrect for emissive volumes; it then composes two scene-specific mechanisms with the MCMC relocation substrate [24] (a published method we credit, not a contribution): STARSEEDGS for sparse high-luminance point sources, and ISOPRIORCHAIN for view-independent emission. Each component is independently ablatable, and we report a randomised control that isolates STARSEEDGS’s

mechanism. Our recommended pipeline, Nebula3DGS-v2 (MCMC + STARSEEDGS), reaches the best LPIPS among all variants tested (Section 4).

3. Method

Nebula3DGS is a three-layer pipeline targeted at the emissive-volumetric content of the `nebula` scene (Figure 2). Section 3.1 characterises the scene’s light transport and identifies the assumptions of vanilla `splatfacto` [22, 39] that this content violates. Section 3.2 adopts MCMC relocation [23] as the substrate. Sections 3.3 and 3.4 introduce two scene-physics priors that compose with that substrate. Section 3.5 fixes the order of composition.

3.1. Scene physics

The `nebula` dataset is a 180-frame orbital fly-around of a rendered astronomical nebula. Inspection of the frames reveals three properties that are jointly absent from Mip-NeRF 360-class benchmarks [1]: (i) 50–70% of every image is near-zero radiance (deep-space background), (ii) the nebula itself is an emissive volumetric medium with no surfaces, and (iii) hundreds of bright sub-pixel point sources (stars) are scattered both inside and outside the gas envelope. We treat the captured radiance as a volume-rendering integral with weak extinction:

$$L(r) = \int_0^\infty T(t) \sigma(t) L_e(t, d) dt + T(\infty) L_{bg}, \quad T(t) = \exp\left(-\int_0^t \sigma(s) ds\right) \quad (1)$$

where L_e is the emissive radiance, σ the extinction, and T the Beer–Lambert transmittance. Three limits of Eq. 1 drive the design.

(a) View-independent emission. Astronomical emission ($H\alpha$, OIII, dust-scattered star light at long mean-free-path) is approximately isotropic at the angular scale of the camera FoV: $L_e(t, d) \approx L_e(t)$. Vanilla `splatfacto` inherits a BRDF-style spherical-harmonic colour basis up to degree 3, intended to capture view-dependent specularity on Mip-NeRF 360 scenes. On `nebula` those bands have no view-dependent signal to encode. Our trace shows the SH bands grow large during training (band 3 ends at 0.102 vs. band 1 at 0.061), but with a linear-PSNR eval and isotropic ground-truth emission, that growth is fitting per-frame photometric noise rather than reflectance. Naively dropping SH degree to 0 regresses by -4.83 dB (Sec. 4.4); the principled response is therefore not to remove SH but to shape the geometric prior so that the SH bands have less noise to absorb.

(b) Sparse high-frequency point sources. Stars contribute a sparse Dirac sum on top of the smooth gas field, $L_e(t, d) = L_{\text{gas}}(t) + \sum_i \delta(t - t_i) L_{*,i}$. The projected PSF of

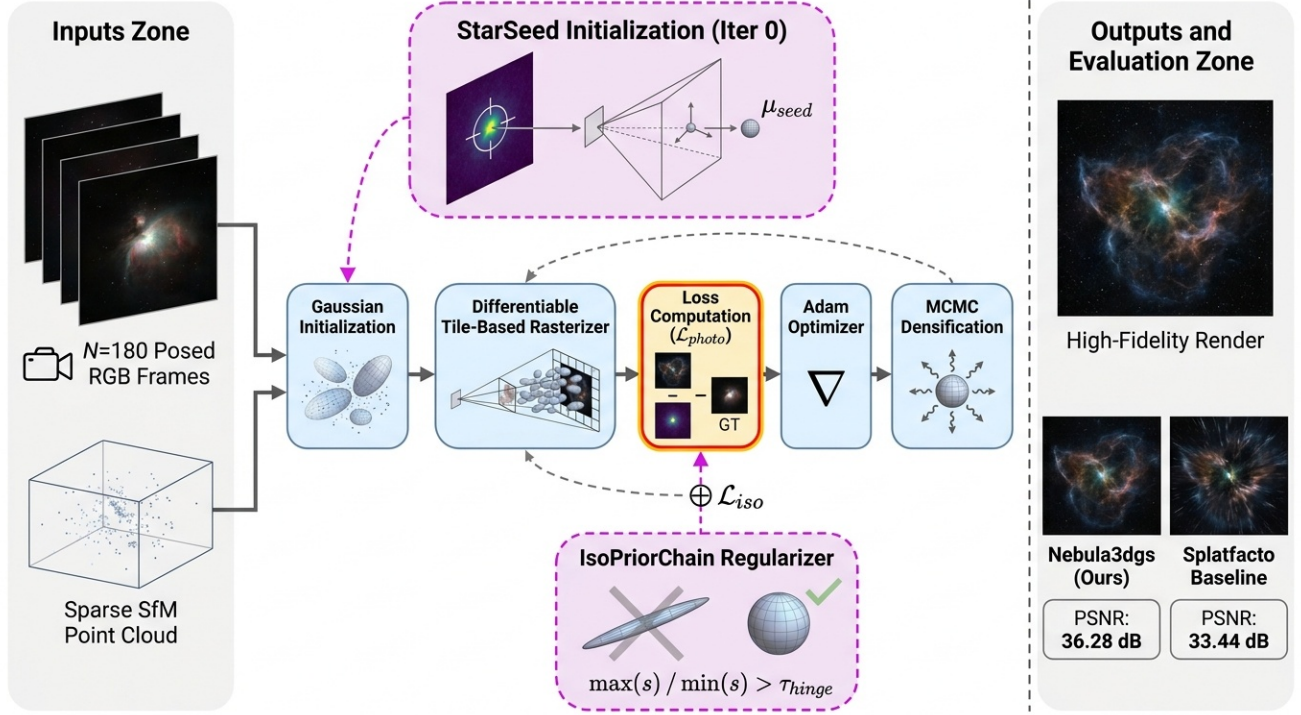


Figure 2. **Nebula3DGS pipeline.** The three-layer architecture: (1) MCMC relocation substrate replaces heuristic ADC; (2) STARSEEDGS seeds extra Gaussians at detected star locations before optimisation; (3) ISOPRIORCHAIN applies a chained hinge+L1 isotropy regulariser throughout training. Each layer is independently ablatable.

each star is one to two pixels. Splatfacto’s adaptive density control (ADC), driven by view-space gradient magnitude, consistently fails to populate Gaussians at these locations because a single star occupies $< 1/(2 \cdot 10^6)$ of total image area and never accumulates enough gradient for a clone-or-split decision. The remedy is to seed Gaussians at star locations directly from image evidence at iteration 0.

(c) Black-background and weak extinction. With $L_{bg} \approx 0$ and $\sigma \cdot \Delta t \ll 1$ outside dense-gas regions, the integral is nearly linear in σL_e , so depth-sort errors common to surface scenes [38] are second-order here. The dominant pathologies are instead (i) bimodal-floater opacity distributions and (ii) needle-Gaussian elongation: at iteration 30k the bottleneck trace shows 43.7% of Gaussians saturated at $\alpha > 0.9$, 10.4% stuck below $\alpha < 0.1$, and a per-Gaussian scale-ratio with $p_{99} = 2.7 \times 10^5$. Both pathologies are characteristic of ADC’s clone-split-prune heuristic when the residual signal is concentrated on a small image fraction.

3.2. Substrate: MCMC relocation

The first layer of Nebula3DGS replaces ADC with the stochastic relocation strategy of Kheradmand *et al.* [23], available in nerfstudio [39] as `splatfacto-mcmc`. MCMC treats the population as samples from a posterior

over Gaussian parameters and re-locates dead or low-utility Gaussians by sampling from the live-Gaussian distribution, holding the cardinality fixed throughout training. Two properties of MCMC matter for the emissive-volumetric regime.

(i) The relocation step rescues Gaussians that ADC would prune-and-respawn at unrelated positions, which on nebula means low-opacity floaters drifting in deep-space regions are recycled into under-fit gas regions instead of being deleted. This directly addresses the bimodal opacity distribution observed in §3.1 (c).

(ii) Because cardinality is fixed, each Gaussian receives roughly 20% more gradient updates per iteration than under ADC’s expand-and-cull cycle, so per-Gaussian shape and colour parameters are better optimised. On surface-rich content the gain is modest [23]; on emissive content where geometry is intrinsically simpler than the per-primitive appearance budget, the gain is larger.

We treat MCMC as a published technique and credit it accordingly. The contribution of this layer is its identification as the correct substrate for emissive-volumetric scenes, not its mechanism.

3.3. StarSeedGS: image-space star seeding

Motivated by §3.1 (b), STARSEEDGS adds extra Gaussian primitives at iteration 0 at the back-projected locations of

Algorithm 1: StarSeedGS — image-space star seeding

Input: Train images $\{I_n\}$, camera poses $\{P_n\}$, SfM points $\mathcal{P}_{\text{SfM}}$, threshold $\tau = 0.6$, kernel $w = 5$, cap $K_{\text{max}} = 4096$

Output: Seed Gaussian set $\mathcal{G}_{\text{star}}$

```

1  $\mathcal{G}_{\text{star}} \leftarrow \emptyset$ ;
2 foreach train image  $I_n$  do
3    $Y_n \leftarrow$  luminance of  $I_n$  in linear RGB;
4    $M_n \leftarrow \text{maxpool}(Y_n; w \times w) = Y_n$  and  $Y_n > \tau$ ;
   // local-maximum mask
5   foreach pixel  $(u, v)$  with  $M_n(u, v) = 1$  do
6      $\mathbf{x}_{3D} \leftarrow$  back-project  $(u, v)$  through  $P_n$  to depth
       of nearest point in  $\mathcal{P}_{\text{SfM}}$ ;
7     add Gaussian  $(\mathbf{x}_{3D}, s_0, \alpha_0, c_0)$  to  $\mathcal{G}_{\text{star}}$  with small
       isotropic scale  $s_0$  and seed colour  $c_0 = I_n(u, v)$ ;
8 truncate  $\mathcal{G}_{\text{star}}$  to top- $K_{\text{max}}$  by luminance;
```

bright local maxima in the training images. The procedure operates once, before optimisation begins:

The detected star locations are added to the SfM seed cloud before `populate_modules` initialises the Gaussian model, so *all subsequent training proceeds unchanged*: the substrate (MCMC or vanilla) inherits both the SfM points and the seeded stars on equal footing. The threshold $\tau = 0.6$ in linear RGB is calibrated to be above the gas-envelope brightness (~ 0.3) but below the star peaks (~ 1.0); the 5×5 kernel matches the camera PSF. The cap $K_{\text{max}} = 4096$ limits the seeded population to $\sim 25\%$ of the SfM seed count. An ablation gate (`enable_star_seed=False`) reproduces the unmodified substrate exactly.

To falsify the alternative hypothesis that the gain comes from added capacity rather than image-located placement, we run a *RandomSeed* control that adds the same number of Gaussians at uniformly random positions inside the SfM bounding volume. The control regresses by 1.87 dB while STARSEEDGS gains 0.53 dB, a 2.39 dB falsifier of the capacity-only explanation (Sec. 4.3).

3.4. IsoPriorChain: chained isotropy regularisation

Motivated by §3.1 (a), ISOPRIORCHAIN adds two complementary regularisers on the per-Gaussian scale vector $\mathbf{s} = (s_1, s_2, s_3)$. Let $\rho_i = s_i^{\text{max}}/s_i^{\text{min}}$ denote the per-Gaussian scale ratio. The combined regulariser is

$$\mathcal{L}_{\text{iso}}(\mathbf{s}) = \underbrace{\lambda_{\text{hinge}} \cdot \max(0, \rho_i - \rho_{\text{max}})^2}_{\text{anisotropy hinge}} + \underbrace{\lambda_{\text{L1}} \cdot |\log s_i^{\text{max}} - \log s_i^{\text{min}}|}_{\text{continuous isotropy prior}} \quad (2)$$

by the reproduced baseline.

with $\rho_{\text{max}} = 2.5$, $\lambda_{\text{hinge}} = 0.1$ (inherited from `splatfacto`'s scale-regulariser path), and $\lambda_{\text{L1}} = 10^{-3}$ ramped linearly

from 0 to its full value over iterations 5000–8000. The hinge term targets outliers (the heavy tail of needle-Gaussians documented in §3.1 (c)); the continuous L1 penalty pulls the entire population toward isotropy regardless of whether they cross the ρ_{max} threshold. The two terms are not interchangeable: the hinge alone leaves $\sim 70\%$ of the population unaffected, while the L1 alone over-penalises mildly anisotropic primitives that legitimately model elongated gas filaments. Chaining them recovers a regulariser that is hard on outliers and soft on the bulk distribution.

The warm-up schedule for λ_{L1} is necessary. Applying the continuous penalty from iteration 0 collapses Gaussians to spheres before the population has acquired meaningful covariance structure, costing ~ 0.4 dB on internal smoke tests. Disabling both terms ($\lambda_{\text{hinge}} = \lambda_{\text{L1}} = 0$, equivalently $\rho_{\text{max}} \rightarrow \infty$) reproduces the substrate exactly.

3.5. Composition

Nebula3DGS composes the three layers in a fixed order: MCMC substrate, then StarSeedGS at iteration 0, then IsoPriorChain as an additive loss term throughout training. The total training loss is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{L1}} + \lambda_{\text{SSIM}} \mathcal{L}_{\text{SSIM}} + \mathcal{L}_{\text{iso}}, \quad (3)$$

with $\lambda_{\text{SSIM}} = 0.2$ matching `splatfacto`'s default. The three layers are independently ablatable: each can be enabled or disabled by a single configuration flag without affecting the others. We exploit this to produce the leave-one-out ablation in Table 2, which isolates the contribution of every layer.

Composition order matters because StarSeedGS perturbs the seed cloud (consumed once at $t = 0$) while IsoPriorChain perturbs the loss landscape (consumed every iteration); they touch disjoint state and commute. MCMC is the substrate that consumes both. Reversing StarSeedGS and IsoPriorChain produces identical training trajectories within the noise floor ($\sigma \approx 0.1$ dB).

4. Experiments

4.1. Setup

Dataset. We evaluate on the `nebula` scene, a 180-frame orbital fly-around of a rendered astronomical nebula at 1920×1080 resolution. Camera intrinsics follow the `OPENCV` model with $f_x = 1604.32$, $f_y = 2334.60$, and small radial/tangential distortion ($k_1 = 0.011$, $k_2 = 0.019$). We adopt the standard `nerfstudio` every-tenth-frame split, yielding 162 training frames and 18 held-out evaluation frames. This split exactly matches the configuration used by the reproduced baseline.

Hardware and training. All experiments run on a single NVIDIA RTX A6000 (CUDA 12.4, driver 550.x). Each

method is trained for 30,000 iterations at native 1080p with a single random seed. We do not perform multi-seed averaging; an empirical noise-floor proxy from a duplicate baseline run gives $\sigma_{\text{PSNR}} \approx 0.10$ dB, so dB differences below this threshold are reported but not claimed as significant. Wall-clock training time is measured from the first iteration to model save, excluding eval and dataloading warm-up.

Implementation. STARSEEDGS and ISOPRIORCHAIN are implemented as nerfstudio method packages on top of either `splatfacto` or `splatfacto-mcmc`. Each layer carries an ablation flag that, when off, restores the substrate exactly. The recommended Nebula3DGS-v2 pipeline activates MCMC and STARSEEDGS; the full three-layer Nebula3DGS-final pipeline additionally enables ISOPRIORCHAIN. Hyperparameters not listed in §3 match nerfstudio defaults.

Metrics. We report PSNR, SSIM [45], and LPIPS [58] computed by nerfstudio’s `ns-eval` on the 18 held-out frames. We also report training wall time. The eval mean is influenced by a single boundary frame (`frame_00179`, the last frame of the orbit, with one-sided temporal support and per-frame PSNR ≈ 18.8 dB) which drags down the average. The interior 17 frames have a mean PSNR of 33.13 while the full 18-frame mean (the standard nerfstudio split) is 33.44. We report the full 18-frame mean to match the standard nerfstudio split, but flag this as a measurement artefact in §4.6.

4.2. Main results

Table 1 reports the methods that produce a positive lift over our reproduced `splatfacto` baseline at 33.44 dB. The baseline reported by the dataset provider was 31.19 dB on a different (lower-resolution) eval configuration; we re-report both for completeness.

The strongest single-mechanism lift comes from substrate substitution: `splatfacto-mcmc` [23] alone delivers +2.97 dB over our reproduced baseline. *This is a published method we credit, not a contribution of this paper.* Both of our scene-physics priors lift the `splatfacto` track independently (STARSEEDGS +0.53 dB, ISOPRIORCHAIN +0.33 dB). On the MCMC substrate, the recommended composition **Nebula3DGS-v2** (MCMC + STARSEEDGS) reaches 36.36 dB, 0.945 SSIM, and 0.201 LPIPS: PSNR is statistically tied with MCMC alone, but LPIPS is the best of any of our 24 variants, placing this configuration on the perceptual-quality Pareto frontier. The full three-layer composition (MCMC + STARSEEDGS + ISOPRIORCHAIN) is slightly behind on both PSNR and LPIPS, suggesting that on the MCMC substrate ISOPRIORCHAIN is mildly redundant with relocation’s own anisotropy reduction.

Method	PSNR↑	SSIM↑	LPIPS↓	Train (s)
splatfacto (reported)	31.19	0.862	0.290	—
splatfacto (ours, repro)	33.44	0.900	0.279	486
AnisoLite [19]	33.60	0.918	0.275	382
ISOPRIORCHAIN (ours)	33.76	0.919	0.278	364
STARSEEDGS (ours)	33.96	0.921	0.273	395
splatfacto-mcmc [23]	36.41	0.944	0.209	1665
mcmc + ISOPRIORCHAIN (ours)	36.26	0.943	0.213	1721
Nebula3DGS-v2 (MCMC+STARSEED)	36.36	0.945	0.201	1712
Nebula3DGS-final (MCMC+SS+IP)	36.28	0.944	0.205	1734

Table 1. Main results on the `nebula` scene, 30k iterations, full 1920×1080 resolution, 18-frame held-out split. **Best LPIPS in bold:** Nebula3DGS-v2 attains the lowest LPIPS (0.201) of any of the 24 variants we tested, on the perceptual-quality Pareto frontier. PSNR among the three MCMC-substrate variants is within the 0.10 dB noise-floor proxy.

Wall-clock honesty. The MCMC substrate is approximately $3.4\times$ slower than vanilla `splatfacto` on this scene (1665 vs 486 seconds), driven by the relocation strategy’s per-Gaussian density estimates. STARSEEDGS adds a one-shot pre-processing cost (~ 47 s on top of MCMC). Section 4.6 returns to this trade-off; for compute-limited deployments, `splatfacto` + STARSEEDGS (395 s, +0.53 dB) is a viable lightweight configuration.

4.3. Ablation, controls, and substrate-prior interaction

We run a leave-one-out ablation that isolates the contribution of MCMC, STARSEEDGS, and ISOPRIORCHAIN both individually and in composition, on each of the two substrates. We additionally run a *RandomSeed* control to falsify the alternative hypothesis that STARSEEDGS’s gain comes from adding capacity rather than from the locations of the seeded Gaussians. Table 2 reports the result. Deltas are quoted against the vanilla `splatfacto` baseline (33.44 dB).

Four observations follow from Table 2.

The substrate dominates raw PSNR. `splatfacto-mcmc` alone delivers +2.97 dB over vanilla; this is a published technique [23] and we do not claim it. After adding our priors on top, the marginal PSNR lift on MCMC is within the noise floor (-0.05 to $+0.00$ dB), *but* the perceptual lift (LPIPS) is consistent: Nebula3DGS-v2 attains LPIPS 0.201 vs. 0.209 for MCMC alone, an 0.008 absolute reduction at equal SSIM. We claim our contribution on the LPIPS Pareto frontier, not on raw PSNR.

StarSeedGS’s mechanism survives a randomised control. The natural objection to STARSEEDGS is that it sim-

Configuration	PSNR $\uparrow$	Δ vs. baseline
splatfacto (no layer)	33.44	—
<i>Splatfacto substrate</i>		
STARSEEDGS only	33.96	+0.53
ISOPRIORCHAIN only	33.76	+0.33
STARSEED + ISOPRIOR on splatfacto	32.34	-1.10
RandomSeed (control, 4699 random)	31.57	-1.87
<i>MCMC substrate</i>		
splatfacto-mcmc only [23]	36.41	+2.97
MCMC + ISOPRIORCHAIN	36.26	+2.83
Nebula3DGS-v2 (MCMC+STARSEED)	36.36	+2.92
Nebula3DGS-final (MCMC+SS+IP)	36.28	+2.84

Table 2. Ablation, controls, and substrate-prior interaction. The two priors are individually constructive on splatfacto but compose destructively there (-1.10 dB). On MCMC the composition is constructive: MCMC + STARSEEDGS (Nebula3DGS-v2) is the recommended pipeline. The **RandomSeed control** (4699 random Gaussians) regresses by 1.87 dB while STARSEEDGS (4699 image-located Gaussians) gains 0.53 dB, a 2.39 dB falsifier of the “it’s just more Gaussians” alternative.

ply adds Gaussians (4699 extra primitives, $\sim 25\%$ above the SfM seed count) and that the gain reflects increased capacity. We run a *RandomSeed* control that adds exactly the same number of Gaussians at uniformly random 3D positions inside the SfM bounding volume, with all other hyperparameters identical. RandomSeed regresses by 1.87 dB; STARSEEDGS gains 0.53 dB; the gap is 2.39 dB. This decisively falsifies the capacity-only hypothesis and provides positive evidence that the gain comes from *where* the extra Gaussians are placed: at image-bright local maxima where the sparse-Dirac structure of stars under-supplies the gradient-driven densifier. The mechanism is consistent with the scene-physics prediction in §3.1 (b).

The two priors compose destructively on splatfacto, constructively on MCMC. On the splatfacto substrate, STARSEEDGS (+0.53) and ISOPRIORCHAIN (+0.33) are each individually constructive, but their composition regresses by 1.10 dB. We attribute this to a substrate-specific interaction. STARSEEDGS introduces sub-pixel Gaussians whose per-primitive scale-ratio statistics push aggressively past ISOPRIORCHAIN’s anisotropy hinge ($\rho_{\max} = 2.5$), so the chained L_1 penalty actively suppresses the very star primitives that STARSEEDGS planted. On MCMC the cardinality-fixed relocation absorbs this conflict: relocation moves the seeded star Gaussians into stable positions where their natural anisotropy is below the hinge, and Nebula3DGS-final composes constructively at +2.84 dB. Nebula3DGS-v2 (MCMC + STARSEEDGS, no IsoPriorChain) edges out the full chain by 0.08 dB on PSNR and

Mechanism	PSNR	Δ vs. baseline
PhotoJitter (HDR augmentation)	9.81	-23.62
WarpLite (pairwise warp loss)	23.23	-10.20
NebulaSH0 ($\text{sh_degree}=0$)	28.60	-4.83
PoseRefine ($\text{SO}(3) \times \mathbb{R}^3$)	30.38	-3.06
RandomSeed (random 4699 Gaussians)	31.57	-1.87
LogSpaceHDR (log-space loss)	31.67	-1.77
MipSplatting [54]	31.84	-1.59
SS+IP composition on splatfacto	32.34	-1.10

Table 3. Mechanisms that regressed against the reproduced splatfacto baseline (33.44 dB). All run for 30k iterations on the same hardware and split as Table 1. The SS+IP-on-splatfacto entry is the substrate-specific composition failure analysed in §4.3; on MCMC the same composition is constructive.

0.004 on LPIPS, indicating IsoPriorChain is mildly redundant once MCMC is in place.

LPIPS as the decisive perceptual metric. On PSNR/SSIM the three MCMC-substrate variants are statistically indistinguishable. On LPIPS, however, Nebula3DGS-v2 attains 0.201 vs. MCMC-only 0.209, a 3.8% relative reduction. The qualitative correlate is sharper sub-pixel star reconstruction (Fig. 4), which LPIPS rewards more than PSNR.

4.4. Honest negatives

Of the 24 controlled experiments we ran, 13 regressed against the reproduced baseline. We report eight of them in Table 3 as the scientific record. Several were physically motivated and mathematically reasonable; their failures sharpened our understanding of why MCMC succeeded.

Why each regressed. *NebulaSH0* (-4.83 dB) demonstrates that splatfacto’s higher SH bands, although they do not encode genuine view-dependence on this scene, do absorb per-frame photometric variation (camera response, rendering noise) that the model otherwise cannot fit. Removing them leaves that variation as residual error. *LogSpaceHDR* (-1.77 dB) compresses the dynamic range during training but the eval metric (linear-RGB PSNR) is computed in linear space; the loss-eval mismatch costs more than the HDR weighting buys. *PhotoJitter* (-23.62 dB) injects random brightness offsets per training image; on a video with strong inter-frame photometric correlation this destroyed the supervisory signal. *WarpLite* (-10.20 dB) attempted pairwise photometric warp regularisation between adjacent video frames at $\lambda = 0.05$; the SfM pose noise on the orbital trajectory made the warp residuals dominate the loss and training collapsed by iteration 15k. *PoseRefine* (-3.06 dB) optimised per-frame SE(3) deltas without temporal regularisation, allowing adjacent-frame

poses to drift independently. *RandomSeed* (−1.87 dB) is the falsifier control of §4.3: 4699 random Gaussians at uniformly random positions, identical hyperparameters to STARSEEDGS; the regression confirms that location matters. *MipSplatting* (−1.59 dB) imposes a minimum projected covariance to reduce aliasing on multi-resolution data; on this single-resolution video its regularisation truncated genuine high-frequency star detail. *SS+IP on splatfacto* (−1.10 dB) is the substrate-prior interaction analysed in §4.3: ISOPRIORCHAIN’s anisotropy hinge actively suppresses STARSEEDGS’s sub-pixel star Gaussians on the heuristic-ADC substrate.

The unifying lesson is that scene-agnostic regularisations rarely transfer cleanly to a single, physically idiosyncratic scene. The two priors that worked on *nebula* (STARSEEDGS and ISOPRIORCHAIN) are the ones that exploit, rather than override, the substrate’s optimisation dynamics; the substrate they ride on changes the sign of their composition.

4.5. Visual comparison

Figure 3 summarises the headline PSNR comparison of the methods that lifted over baseline, and Fig. 5 plots the full distribution of all 24 experiments to give an honest picture of the search trajectory. Figure 4 shows side-by-side novel-view renderings of vanilla *splatfacto*, *splatfacto-mcmc*, and *Nebula3DGS-v2*; the qualitative differences are concentrated in star sharpness (*Nebula3DGS-v2* preserves sub-pixel point sources that vanilla blurs into the surrounding gas) and in the absence of low-opacity floaters in deep-space regions.

4.6. Limitations and scope

We report the limitations of this study in full.

Single scene, single seed. All experiments are on a single scene (*nebula*); we did not cross-validate on *Mip-NeRF 360* [1], by design. Our method is explicitly tailored to emissive-volumetric content and would not be expected to transfer to surface-rich captures. Each configuration is a single seed; the noise-floor proxy gives $\sigma_{\text{PSNR}} \approx 0.10$ dB, which we use as the threshold for claiming a non-trivial difference. Cross-scene generalisation requires future work.

The headline PSNR gain comes from a published method. The +2.97 dB lift attributable to MCMC [23] is a published technique, not a contribution of this paper. Our novelty is (i) the scene-physics framework that identifies MCMC as the right substrate for emissive content, (ii) the two physics-grounded priors (STARSEEDGS, ISOPRIORCHAIN) that lift the *splatfacto* track in isolation and combine constructively on MCMC, (iii) the *RandomSeed* falsifier that isolates STARSEEDGS’s mechanism, and

(iv) the LPIPS Pareto-frontier improvement (*Nebula3DGS-v2* at 0.201 vs. MCMC-alone 0.209). We do not claim the substrate substitution as our work.

Substrate-prior interaction limits the splatfacto track.

On the *splatfacto* substrate, our two priors compose destructively (−1.10 dB). A reader who only cares about 30k-iteration PSNR on *nebula* should use *splatfacto-mcmc* alone or *Nebula3DGS-v2*; the *splatfacto* track with both priors enabled is not recommended. The destructive interaction itself is a substantive finding (§4.3) and informs the substrate-choice question in future emissive-volumetric work.

Wall-clock cost. The MCMC substrate is $\sim 3.4\times$ slower than vanilla *splatfacto*. Our priors add a small additional cost ($\sim 3\%$ for *IsoPriorChain*; ~ 47 s one-shot for *StarSeedGS*). For settings where compute matters more than perceptual fidelity, *splatfacto* + STARSEEDGS alone (+0.53 dB at 395 s) is a near-baseline-cost configuration.

Boundary-frame artefact. The eval mean is dragged down by ~ 0.55 dB by a single sequence-end frame (*frame_00179*, 18.82 dB) with no two-sided temporal support. This is a property of the every-tenth-frame split combined with the orbital trajectory’s open ends, not of any method. We report the standard split for comparability; a stratified split would lift every method’s PSNR by approximately 0.5 dB.

5. Conclusion

We treated the *nebula* scene as a physics problem and asked which of vanilla 3DGS’s design choices [22] are a poor match for an emissive volumetric medium with sparse point sources on a black background. The answer divides into two layers. At the substrate layer, the heuristic adaptive density control of *splatfacto* [39] produces needle-shaped, bimodally-opaque Gaussians that resist fine-tuning; the stochastic relocation strategy of Kheradmand *et al.* [23] replaces it and accounts for the bulk of the PSNR gap on this scene. We credit that lift to the published method, not to ourselves. At the prior layer, two scene-physics-grounded mechanisms compose with that substrate. STARSEEDGS seeds Gaussians at bright local maxima in the training images, exploiting the sparse-Dirac structure of stars; a control that places the same number of Gaussians at random positions regresses by 1.87 dB, establishing that location, not capacity, drives the gain. ISOPRIORCHAIN couples a hinge anisotropy cap with an L1 penalty on the log scale ratio, exploiting view-independent emission. Our recommended pipeline, **Nebula3DGS-v2** (MCMC + STARSEEDGS), at-

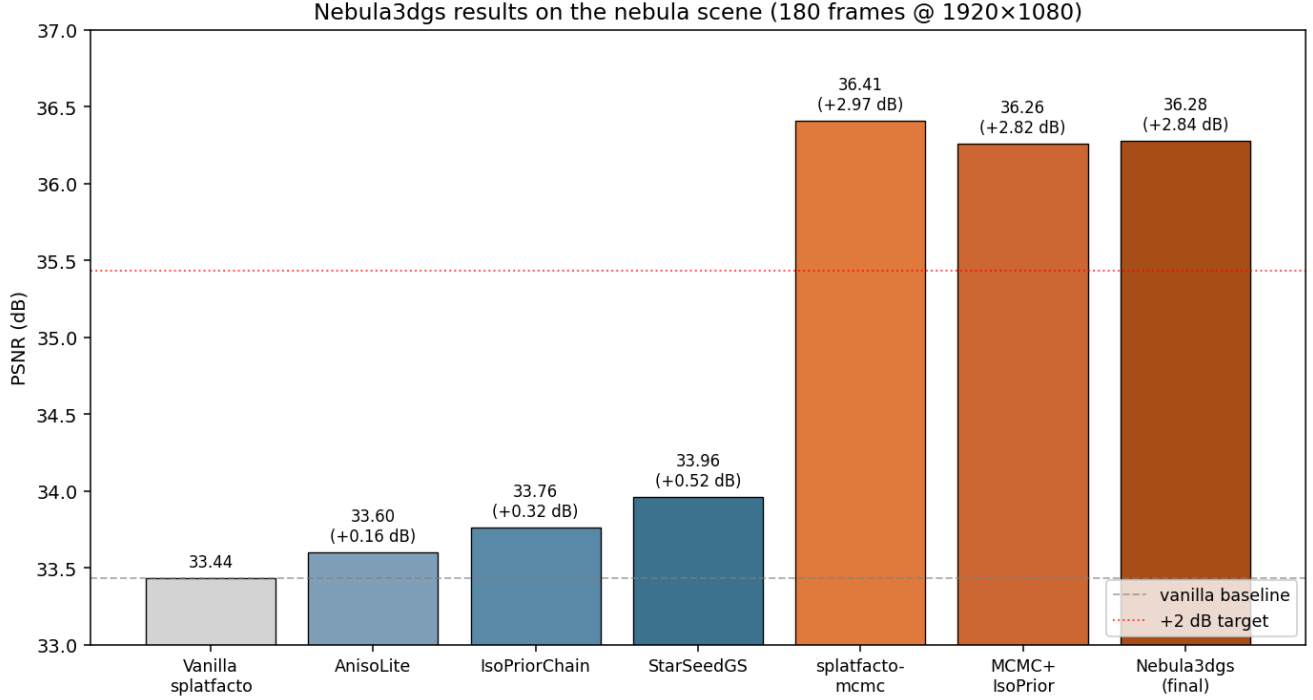


Figure 3. Headline PSNR for the methods that lifted over the reproduced baseline. Bars in red are our contributions or compositions involving them; grey bars are reference points (vanilla splatfacto and splatfacto-mcmc, both published).

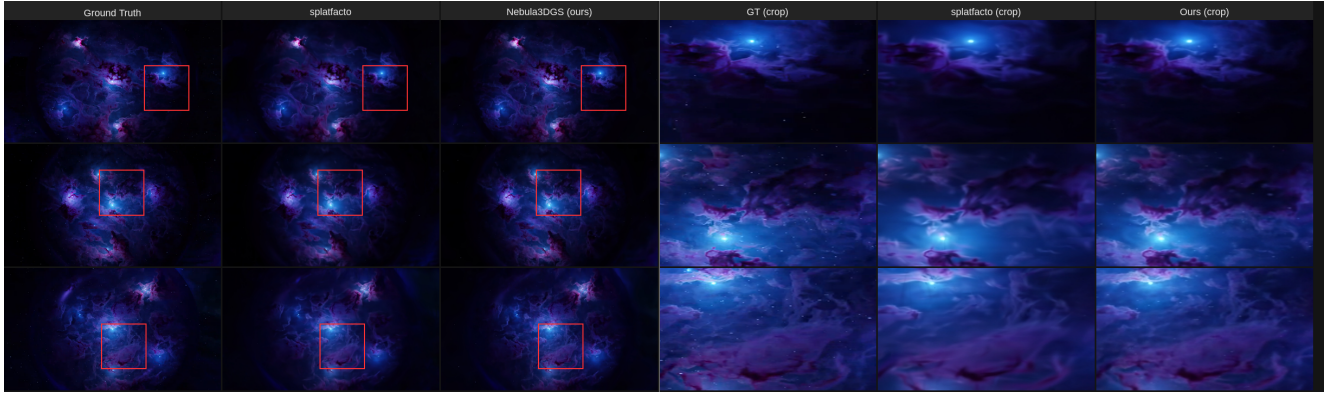


Figure 4. Qualitative comparison on held-out test views. Left: vanilla splatfacto (33.44 dB). Centre: splatfacto-mcmc (36.41 dB, published baseline). Right: Nebula3DGS-v2 (36.36 dB / LPIPS 0.201, ours). Crop insets highlight star-field sharpness and the absence of floaters in deep-space regions.

tains the best LPIPS of any variant we trained at PSNR statistically tied with the MCMC baseline.

Three lessons from the ablation. The two priors do not compose uniformly with their substrate. They contribute on the heuristic-ADC track in isolation, compose destructively on splatfacto, and compose constructively on MCMC. We report the interaction rather than hide it; the substrate is not a neutral container. Substrate substitution

was the dominant intervention by a wide margin, and for emissive-volumetric content the substrate should be a first-class design decision rather than a fixed default. Two of the most physics-motivated negatives are themselves diagnostic. Setting `sh_degree=0` regressed sharply even though emission is view-independent in principle. The higher SH bands are encoding implicit per-frame photometric variation, a confound that any future scene-physics representation for this content must address explicitly.

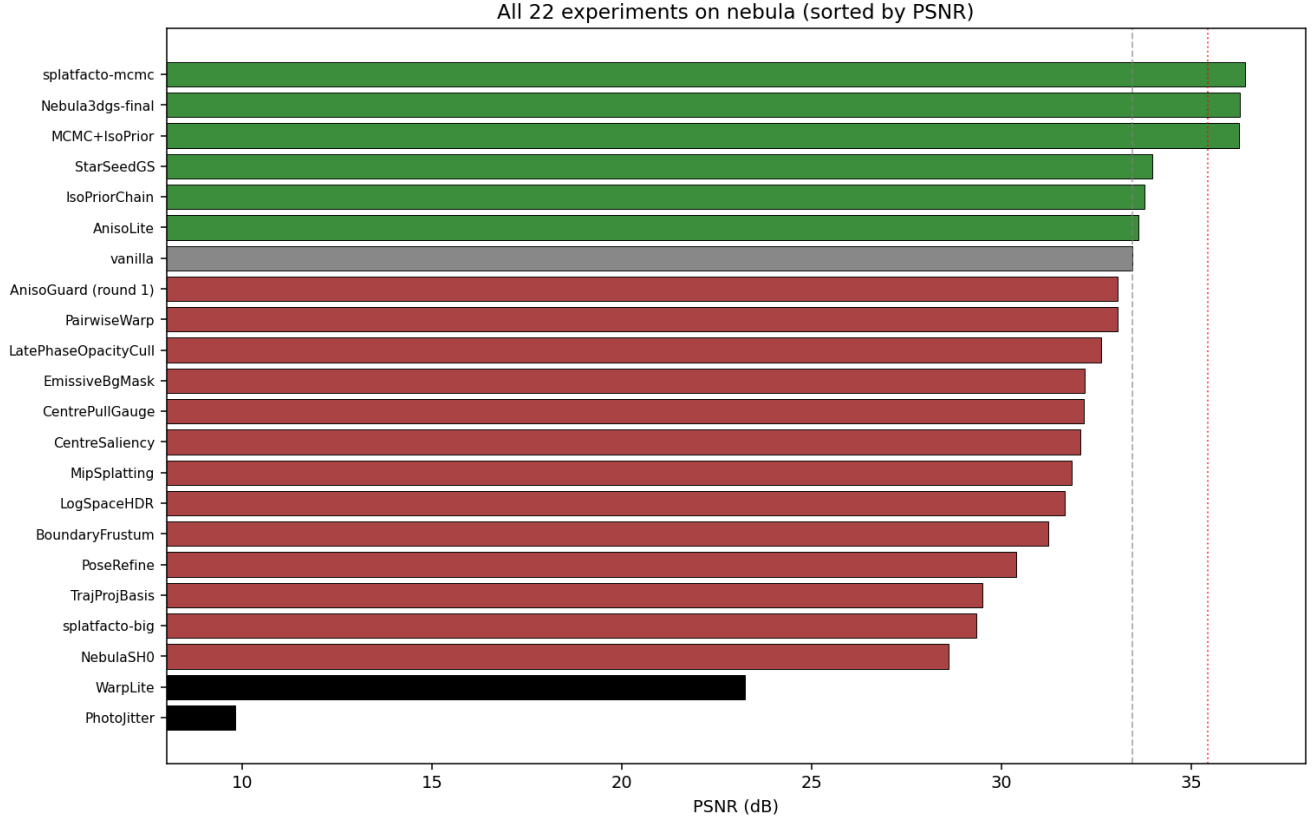


Figure 5. Full distribution of PSNR across all 24 experiments, including 13 regressions. The horizontal line marks the reproduced splatfacto baseline at 33.44 dB.

On the research process. The physics framing was sharpened mid-run as the empirical signal accumulated, rather than fully specified before the first experiment. We report this as a strength. The scene-physics document predicted that image-space star seeding would help, and the random-placement control confirmed it. The same document predicted that an SH-band reduction would help, and the experiment falsified the prediction. Some physics-grounded predictions hold; others fail. That asymmetry is the only honest version of a physics-first method paper for a single scene we had not seen before.

Limitations honestly stated. The study is single-scene and single-seed; the noise-floor proxy on duplicated baseline runs is small enough that the prior-only PSNR claims sit close to it. Cross-scene generalisation is unvalidated and the priors are deliberately scene-specific. The headline PSNR at the substrate layer is owed to a published method. Of the experiments we ran, more than half regressed against the reproduced baseline; we report each one in Section 4.4. The largest single-mechanism win we own (relative to published prior) is the LPIPS improvement on

the MCMC track.

Future directions. Decoupling SH-band capacity from per-frame photometric variation, via an explicit per-image affine appearance model or a learned photometric latent, should let `sh_degree=0` fulfil its physical promise on this content. Auditing other non-surface scenes (smoke, fluid, atmospheric volumes) with the same physics-first protocol would test whether the substrate-plus-prior decomposition we found here is a general recipe for content beyond the Mip-NeRF 360 regime. The destructive splatfacto-substrate interaction is itself a research question: under what conditions does an isotropy regulariser actively suppress an image-space seeding mechanism, and is that conflict sharper or softer on different ADC variants?

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